Physics Isomorphic Questions

Vectors and One-Dimensional Kinematics

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A survey drone starts at the origin of a coordinate grid and detects a beacon at position $\vec{r} = -7\hat{i}+24\hat{j}$, where distances are measured in kilometers. Which of the following statements is correct?

A. The distance from the drone to the beacon is 31 km.
This comes from adding the magnitudes of the components, $7 + 24$, instead of finding the vector magnitude. Distance from the origin must be computed with $|\vec{r}| = \sqrt{x^2 + y^2}$, not by simple addition.

B. The distance from the drone to the beacon is 25 km.
The beacon's position vector has components -7 km and 24 km, so its distance from the origin is the magnitude $|\vec{r}| = \sqrt{(-7)^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25\,\text{km}$. The sign of the x-component affects direction but not the distance, so the correct choice is 25 km.

C. The distance from the drone to the beacon is 24 km.
This treats only the vertical component as the full distance. Both the x- and y-components contribute to the magnitude of the position vector.

D. The distance from the drone to the beacon is 7 km.
This uses only the horizontal component and ignores the vertical displacement. A vector's magnitude must include all perpendicular components through the Pythagorean relation.

Answer: *B*

Q2. The position of a particle moving along the x-axis is given by $\vec{s}(t) = \big((0.80\,\text{m/s}^3)t^3 - (1.90\,\text{m/s}^2)t^2\big)\hat{i}$. Which of the following statements is correct?

A. The magnitude of the acceleration at $t = 0$ is $3.8\,\text{m/s}^2$.
Differentiate twice: $\vec{v}(t) = \big(2.40t^2 - 3.80t\big)\hat{i}$ and $\vec{a}(t) = \big(4.80t - 3.80\big)\hat{i}$. At $t = 0$, the acceleration is $-3.80\hat{i}\,\text{m/s}^2$, so its magnitude is $3.8\,\text{m/s}^2$.

B. This motion has constant acceleration.
This error comes from overlooking the cubic term in position. Since $\vec{a}(t) = \big(4.80t - 3.80\big)\hat{i}$, the acceleration depends on time and is not constant.

C. The magnitude of the acceleration is $1.9\,\text{m/s}^2$.
This confuses the coefficient of the t^2 term in displacement with acceleration itself. Acceleration must be obtained from the second derivative of the displacement function.

D. The magnitude of the acceleration at $t = 0$ is $0\,\text{m/s}^2$.
This mistake assumes the acceleration must vanish initially. Evaluating $\vec{a}(t) = \big(4.80t - 3.80\big)\hat{i}$ at $t = 0$ gives a nonzero value.

Answer: *A*

Q3. A robot moves from position $\vec{r}_A = 1\hat{i} - 2\hat{j}$ m to position $\vec{r}_B = 7\hat{i} + 6\hat{j}$ m in 5.0 s. Which of the following statements is correct?

A. The robot must have traveled a distance of 10 m.
The value 10 m is the magnitude of the displacement, not necessarily the path length traveled. The actual distance traveled depends on the path and cannot be determined from only the initial and final positions.

B. The robot's speed at $t = 0$ must have been 2.0 m/s.
This comes from dividing displacement magnitude by elapsed time and interpreting that as the initial speed. Endpoint positions over a time interval do not uniquely determine the speed at the starting instant.

C. The magnitude of the displacement is 10 m.
The displacement is $\Delta\vec{r} = \vec{r}_B - \vec{r}_A = (7-1)\hat{i} + \big(6-(-2)\big)\hat{j} = 6\hat{i} + 8\hat{j}$ m. Its magnitude is $|\Delta\vec{r}| = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$ m, so this is the only statement that must be true.

D. The robot's acceleration at $t = 0$ must have been 0.40 m/s².
This incorrectly assumes acceleration can be inferred from displacement and total time alone. Without more information about the motion or velocity, the initial acceleration is not determined.

Answer: *C*